Priority of operations with positive and negative integers, decimals and fractions

Task A: Reviewing priority of operations

1) Work out the following using the priority of operations, giving your answer in its simplest form where possible.

a) $\frac{9}{10} + \underbrace{\frac{3}{4} \times \frac{2}{5}}$

$\frac{9}{10} + \frac{6}{20} =$

$\frac{9}{10} + \frac{3}{10} =$

$\frac{12}{10} = \frac{6}{5} = 1\frac{1}{5}$

b) $\frac{11}{15} - \underbrace{\frac{3}{5} \div \frac{9}{10}}$

$\frac{11}{15} - \frac{3}{5} \times \frac{10}{9}$

$\frac{11}{15} - \frac{30}{45} =$

$\frac{11}{15} - \frac{10}{15} = \frac{1}{15}$

c) $\underbrace{\frac{9}{10} \times \frac{1}{2}} - \underbrace{\frac{3}{5} \div \frac{4}{5}}$

$\frac{9}{20} - \frac{3}{5} \times \frac{5}{4}$

$\frac{9}{20} - \frac{15}{20} =$

$\left(-\frac{6}{20}\right) = \left(-\frac{3}{10}\right)$

2) Work out the following using the priority of operations, giving your answer in its simplest form and a mixed number where possible.

a) $2\frac{1}{2} + 3\frac{2}{3} \times \frac{3}{4}$

$\frac{5}{2} + \frac{11}{3} \times \frac{3}{4}$

$\frac{5}{2} + \frac{33}{12}$

$\frac{30}{12} + \frac{33}{12} = \frac{63}{12} = \frac{21}{4} = 5\frac{1}{4}$

b) $2\frac{1}{4} \times \frac{4}{5} - \frac{3}{5} \div 1\frac{3}{10}$

$\frac{9}{4} \times \frac{4}{5} - \frac{3}{5} \div \frac{13}{10}$

$\frac{36}{20} - \frac{3}{5} \times \frac{10}{13}$

$\frac{36}{20} - \frac{30}{65} = \frac{9}{5} - \frac{6}{13}$

$\frac{117}{65} - \frac{30}{65} = \frac{87}{65} = 1\frac{22}{65}$

Task B: Powers and roots of fractions

1) Fill in the blanks to complete the calculation

$$\frac{3}{5} - \frac{4}{5} \times \frac{2}{3} + \left(\frac{3}{5}\right)^2$$

$$\frac{3}{5} - \frac{4}{5} \times \frac{2}{3} + \boxed{\frac{9}{25}}$$

$$\frac{3}{5} - \boxed{\frac{8}{15}} + \boxed{\frac{9}{25}}$$

$$\frac{3}{5} + \boxed{\frac{9}{25}} + \left(-\boxed{\frac{8}{15}}\right)$$

$$\frac{\boxed{90}}{150} + \frac{\boxed{54}}{150} + \left(-\frac{\boxed{80}}{150}\right) = \boxed{\frac{64}{150}} = \boxed{\frac{32}{75}}$$

2) Work out the answer to the following, giving your answer in its simplest form.

$$\frac{9}{10} \times \frac{4}{5} - \frac{2}{3} \times \sqrt[3]{\frac{8}{27}}$$

$$= \frac{9}{10} \times \frac{4}{5} - \frac{2}{3} \times \frac{2}{3}$$

$$= \frac{36}{50} - \frac{4}{9}$$

$$= \frac{324}{450} - \frac{200}{450}$$

$$= \frac{124}{450}$$

$$= \frac{62}{225}$$

3) Lucas has made an error.

Find the error and correctly complete the calculation

$$\frac{2}{3} + \sqrt{6\frac{1}{4}} \times \frac{3}{5}$$

$$\frac{2}{3} + \sqrt{6} + \frac{1}{2} \times \frac{3}{5}$$

$$\frac{2}{3} + \sqrt{6} + \frac{3}{10}$$

$$\sqrt{6} + \frac{2}{3} + \frac{3}{10} =$$

$$\sqrt{6} + \frac{20}{30} + \frac{9}{30} = \sqrt{6}\ \frac{29}{30}$$

Lucas needed to convert the mixed number into an improper fraction first

$$\frac{2}{3} + \sqrt{6\frac{1}{4}} \times \frac{3}{5}$$

$$\frac{2}{3} + \sqrt{\frac{25}{4}} \times \frac{3}{5}$$

$$\frac{2}{3} + \frac{5}{2} \times \frac{3}{5}$$

$$\frac{2}{3} + \frac{15}{10} =$$

$$\frac{20}{30} + \frac{45}{30} = \frac{65}{30} = \frac{13}{6}$$

Task C: Powers and roots of fractions

1) Fill in the blanks to complete the calculation

$$2 - \left(\frac{2}{5}\right)^2 \times \sqrt{\frac{49}{100}}$$

$$2 - \boxed{\frac{4}{25}} \times \boxed{\frac{7}{10}}$$

$$2 - \frac{\boxed{28}}{250}$$

$$\frac{\boxed{500}}{250} - \frac{\boxed{28}}{250} = \frac{\boxed{472}}{250} = \frac{\boxed{236}}{125} = 1\frac{\boxed{111}}{125}$$

2) Work out the answer to the following, ensuring to give your answer as a mixed number in its simplest form

a) $\frac{9}{10} + (2.4 \div 0.8) \times \frac{2}{3}$

$$= \frac{9}{10} + \frac{2.4}{0.8} \times \frac{2}{3}$$

$$= \frac{9}{10} + \frac{24}{8} \times \frac{2}{3}$$

$$= \frac{9}{10} + \frac{48}{24} = \frac{9}{10} + 2 = 2\frac{9}{10}$$

b) $15 + (3 \div 5) \times (0.3 \div 4)$

$$= 15 + \frac{3}{5} \times \frac{0.3}{4}$$

$$= 15 + \frac{3}{5} \times \frac{3}{40}$$

$$= 15 + \frac{9}{200}$$

$$= 15\frac{9}{200}$$

3) Work out the answer to the following, ensuring to give your answer as a mixed number in its simplest form

a) $2 - \frac{3}{4} \times \sqrt{\frac{49}{81}}$

$$2 - \frac{3}{4} \times \frac{7}{9}$$

$$2 - \frac{21}{36}$$

$$\frac{72}{36} - \frac{21}{36} = \frac{51}{36} = \frac{17}{12} = 1\frac{5}{12}$$

b) $\left(\frac{9}{10}\right)^2 - (2.5 \div 0.5) \times 2$

$$\frac{81}{100} - \frac{2.5}{0.5} \times 2$$

$$\frac{81}{100} - \frac{25}{5} \times 2$$

$$\frac{81}{100} - 5 \times 2$$

$$\frac{81}{100} - 10 = -9\frac{19}{100}$$